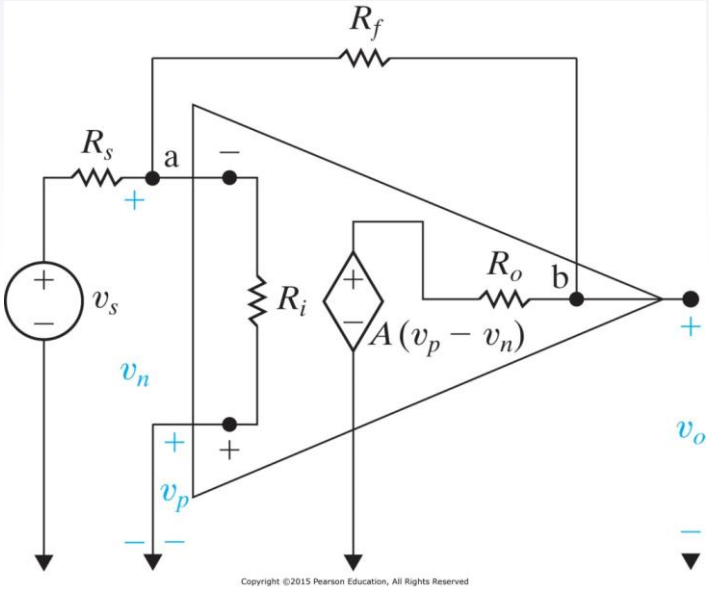
EE2001 Electrical Circuits- I:

Continuation of the topic:
The Operational Amplifier

18 December 2023

Inverting Amplifier Circuit with the More Realistic Op-Amp Model



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- Note that as $R_o \to 0$, $R_i \to \infty$, and $A \to \infty$ we have,

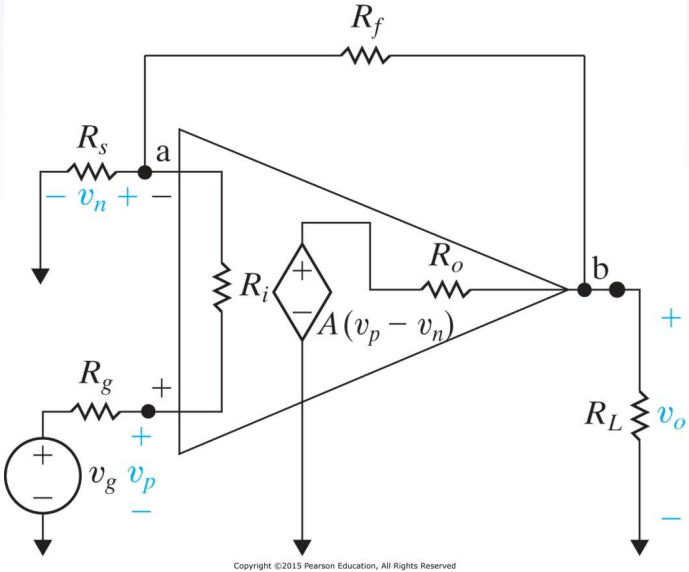
$$v_o \to \frac{-A}{\frac{R_s}{R_f}A} v_s = -\frac{R_f}{R_s} v_s$$

- In case there is a load resistor R_L connected to the output of the op-amp, it needs to be considered as well and the output voltage equation becomes.

$$v_o = \frac{-A + \frac{R_o}{R_f}}{\frac{R_s}{R_f}\left(1 + A + \frac{R_o}{R_i} + \frac{R_o}{R_L}\right) + \left(1 + \frac{R_o}{R_L}\right)\left(\frac{R_s}{R_i} + 1\right) + \frac{R_o}{R_f}} v_s$$

Non-Inverting Amplifier Circuit with the More Realistic Op-Amp Model

- Consider the non-inverting amplifier



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$$\frac{v_n}{R_s} + \frac{v_n - v_p}{R_i} + \frac{v_n - v_o}{R_f} = 0$$

$$\frac{v_o - v_n}{R_f} + \frac{v_o - A(v_p - v_n)}{R_o} + \frac{v_o}{R_L} = 0$$

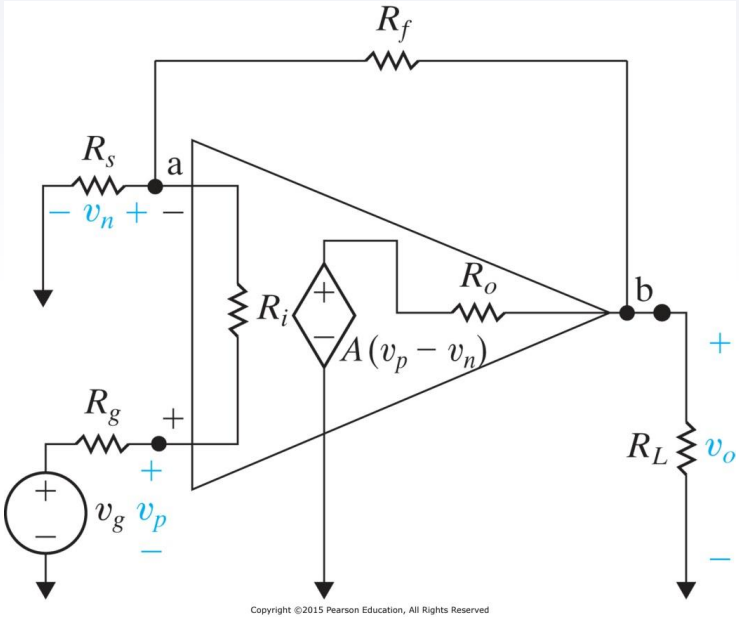
$$\frac{v_p - v_g}{R_g} = \frac{v_n - v_p}{R_i}$$

- Solving these equations for v_o, one obtains,

$$v_o = \frac{\left[(R_f + R_s) + \left(\frac{R_s R_o}{A R_i}\right)\right]}{R_s + \frac{R_o}{A}(1 + K_r) + \frac{R_f R_s + (R_f + R_s)(R_i + R_g)}{A R_i}} v_g$$

$$K_r = \frac{R_s + R_g}{R_i} + \frac{R_f + R_s}{R_L} + \frac{R_f R_s + R_f R_g + R_g R_s}{R_i R_L}$$

Non-Inverting Amplifier Circuit with the More Realistic Op-Amp Model



- Note that as $R_o \to 0$, $R_i \to \infty$, and $A \to \infty$ we have,

$$v_o \to \left(1 + \frac{R_f}{R_s}\right) v_g$$

- In case there is a load resistor R_L is omitted from the output, the output voltage equation becomes.

$$v_o = \frac{\left[(R_f + R_s) + \left(\frac{R_s R_o}{A R_i}\right)\right]}{R_s + \frac{R_o}{A}\left(1 + \frac{R_s + R_g}{R_i}\right) + \frac{R_f R_s + (R_f + R_s)(R_i + R_g)}{A R_i}} v_g$$

EE2001 Electrical Circuits- I:

Continuation of the Topic:

Inductance, Capacitance and Mutual Inductance

23 December 2020

Mutual Inductance

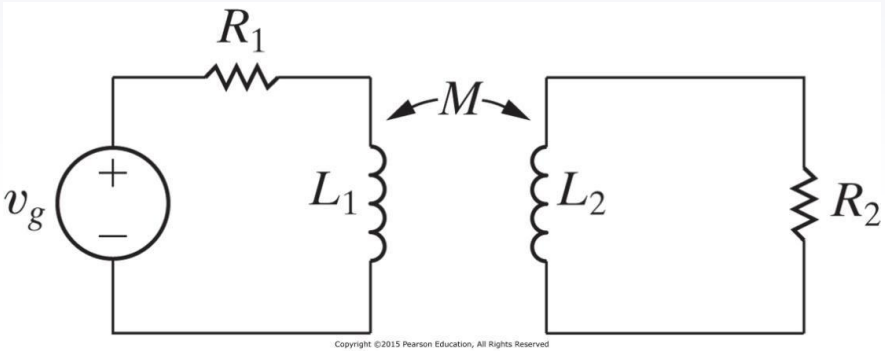
- Recall that the inductance in an inductor relates the voltage to the change in the current.

  - This type of inductance is called self-inductance.

- When two (or more) circuits (or coils) are magnetically coupled -linked by a magnetic field- (i.e., one circuit/coil is situated within the magnetic field induced by the current in the other circuit/coil), change in the current in one of the circuits/coils can induce voltage in the other circuit/coil as well.
- The parameter which relates the voltage induced in one coil/circuit due to the time-varying current in the other (magnetically coupled) circuit/coil is referred to as mutual inductance.

Mutual Inductance

- Coils with mutual inductance can be represented in the form

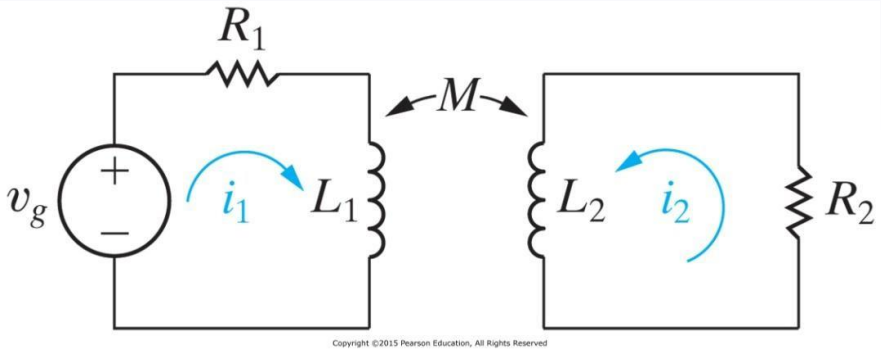


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- L_1, L_2 -> self inductances of the two coils
- M -> mutual inductance between the two coils
- Note that the two coils in the above circuit are coupled only magnetically - there is no direct electrical coupling between the coils.

Mutual Inductance

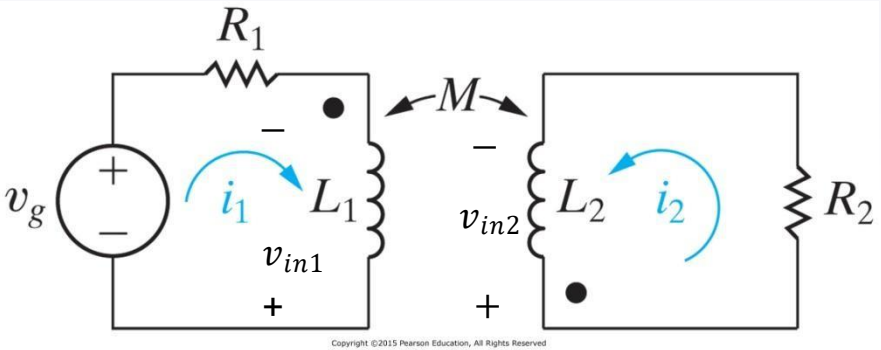
- The current-voltage characteristics of two magnetically coupled coils can be described as follows.



- First define the reference directions of the currents through the coils.
- The polarity of the voltages induced in the coils due to the currents through them are determined using a “dot” system.

Mutual Inductance

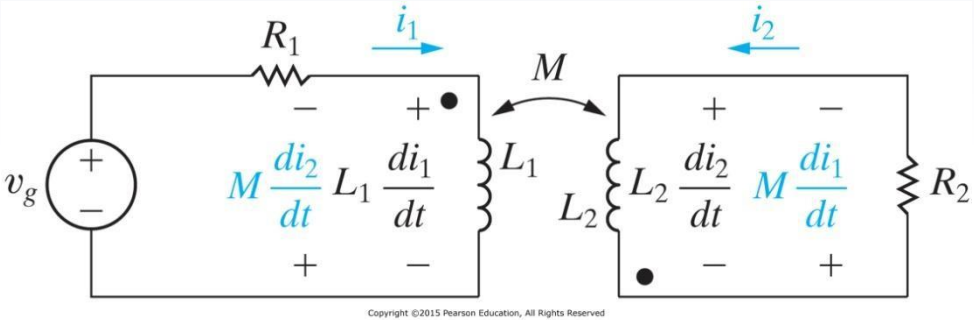
- The dots carry information about the polarity of the mutually induced voltage.



- When the reference direction of the current enters a coil from the its dotter terminal the voltage it induces in the other coil is higher (i.e., has its plus side) at its dotted terminal. Otherwise, it induces a voltage which is lower (i.e., has minus side) at the dotted terminal.

Mutual Inductance

- Given the below reference directions of the currents and the dot marks of the terminals



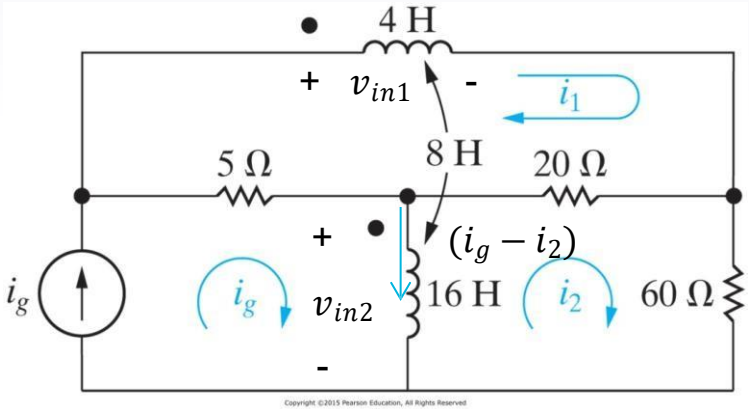
- One can write the current-voltage equations as

$$-v_g + i_1 R_1 + L_1 \frac{di_1}{dt} - M \frac{di_2}{dt} = 0$$

$$i_2 R_2 + L_2 \frac{di_2}{dt} - M \frac{di_1}{dt} = 0$$

Mutual Inductance

- Usually the mesh-current method is used to analyze circuits containing mutual inductance.
- Example: Write a set of mesh-current equations that describe the circuit in terms of currents i_1 and i_2.



Summing the voltages around the i_1 mesh:

$$4\frac{di_1}{dt} + 8\frac{d}{dt}(i_g - i_2) + 20(i_1 - i_2) + 5(i_1 - i_g) = 0$$

The i_2 mesh equation is:

$$20(i_2 - i_1) + 60i_2 + 16\frac{d}{dt}(i_2 - i_g) - 8\frac{di_1}{dt} = 0$$

The voltage across the 4H coil due to the current $(i_g - i_2)$, that is , $8\frac{d}{d}(i_g - i_2)$ is a voltage drop in the direction of i_1.

The voltage induced in the 16 H coil by the current i_1, that is, $8\frac{di_1}{dt}$, is a voltage rise in the direction of i_2.

Mutual Inductance

- In circuits with linear coupling the mutual inductance can be expressed as

$$M = k\sqrt{L_1 L_2}$$

- The coefficient k is called the coupling coefficient.
- The coupling coefficient satisfies $0 \leq k \leq 1$.
- Larger the coupling coefficient, better the coupling between the two coils.
- Note that we have always

$$M \leq \sqrt{L_1 L_2}$$

Mutual Inductance

- The energy stored in magnetically coupled coils can be expressed as

$$w = \frac{1}{2} L_1 {i_1}^2 + \frac{1}{2} L_2 {i_2}^2 \pm M i_1 i_2$$

- The sign in the last term depends on the direction of the currents .

  - If both the currents i_1 and i_2 are entering or leaving the coils from the dotted terminal the sign is positive. Otherwise it is negative.

EE2001 Electrical Circuits- I:

Continuation of the Topic:

Response of First-Order RL and RC Circuits

28 December 2020

Outline



Continuation of the topic:

4. The Step and the Total Responses of RL and RC Circuits

5. Sequential Switching

6. Unbounded Response

The Step and the Total Responses of RL and RC Circuits

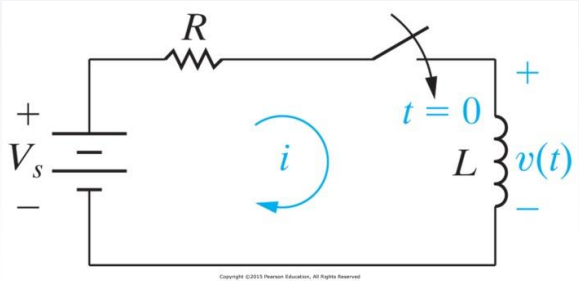
4

- The response of a circuit/system to a sudden application of an external constant excitation (i.e., a constant current or a constant voltage source) is referred to as the step response of the circuit/system.
- For the step response, the initial conditions in the circuit/system are assumed to be zero.
- The response of a circuit/system to a sudden application of an external excitation when there are also non-zero initial conditions is referred to as the total response of the circuit/system.
- Therefore, the total response is the natural response and the step response (or the response to another input) together.

The Step and the Total Responses of RL and RC Circuits

4

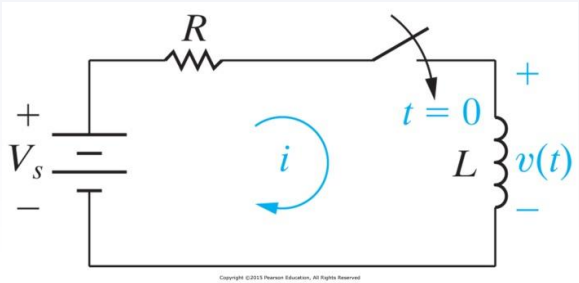
- Consider the circuit



- If $i(0) = 0$, step response.
- If $i(0) = I_0 \neq 0$, total response.
- We can write a loop (KVL) equation to solve for the response: $-V_s + Ri + L\frac{di}{dt} = 0$

The Step and the Total Responses of RL and RC Circuits

4



- If $i(0) = 0$ (step response)

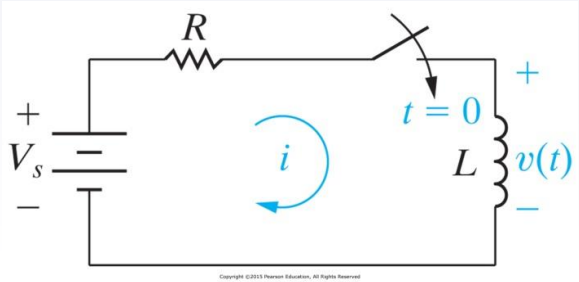
$$i(t) = \frac{V_s}{R} - \frac{V_s}{R} e^{-\frac{R}{L}t}$$

- If $i(0) = I_0 \neq 0$ (total response)

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right) e^{-\frac{R}{L}t}$$

The Step and the Total Responses of RL and RC Circuits

- Consider the total response



$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R}\right)e^{-\frac{R}{L}t}$$

$$v(t) = (V_s - I_0 R)e^{-\frac{R}{L}t}$$

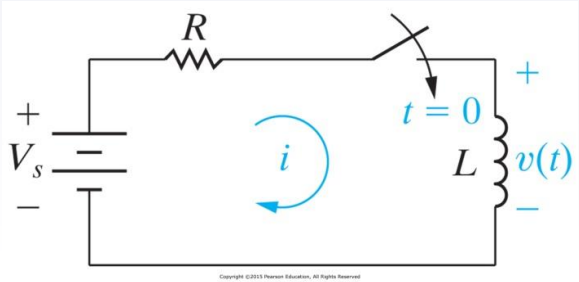
- It can be rewritten in the form

$$i(t) = \underbrace{I_0 e^{-\frac{R}{L}t}}_{\text{natural response}} + \underbrace{\frac{V_s}{R}\left(1 - e^{-\frac{R}{L}t}\right)}_{\text{step response}}$$

- The first part is the response due to the initial conditions (natural response)
- The second part is the response due to the input (step response)

The Step and the Total Responses of RL and RC Circuits

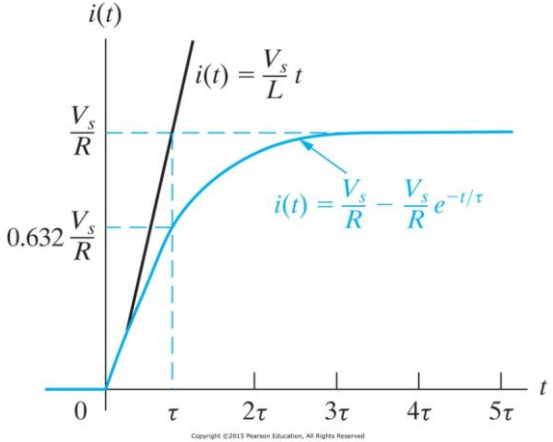
- Consider the step response



$$i(t) = \frac{V_s}{R} - \frac{V_s}{R} e^{-\frac{R}{L}t}$$

$$v(t) = V_s e^{-\frac{R}{L}t}$$

- The plot of the response is in the form



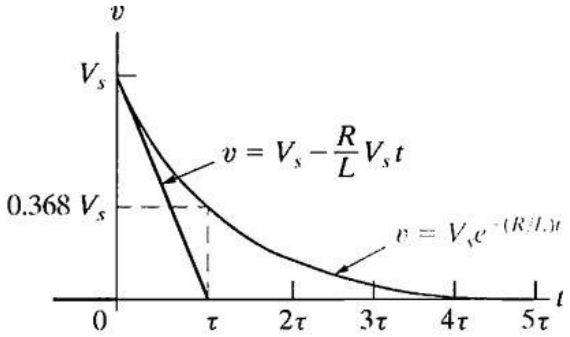


Figure 7.18 ▲ Inductor voltage versus time.

The Step and the Total Responses of RL and RC Circuits

4

- General solution methodology:

$$\text{the unknown variable as a function of time} = \text{the final vaule of the variable} + [\text{the initial value of the variable} - \text{the final value of the variable}] \times e^{\frac{-[t-(\text{time of switching})]}{\text{time constant}}}$$

1- Identify the variable of interest for the circuit.

- For RC circuits, it is best to choose capacitive voltage
- For RL circuits, it is best to choose inductive current

2- Determine the initial value of the variable (the value at t_0).

For continuous variables (capacitive voltage or inductive current), it is not necessary to distinguish between $t = t_0^-$ and $t = t_0^+$.

If you choose another variable, you need to remember that its inital value is defined at $t = t_0^+$

3- Calculate the final value of the variable

4- Calculate the time constant for the circuit

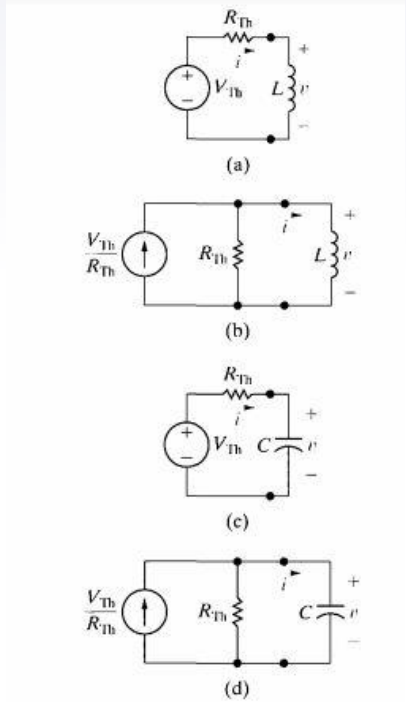


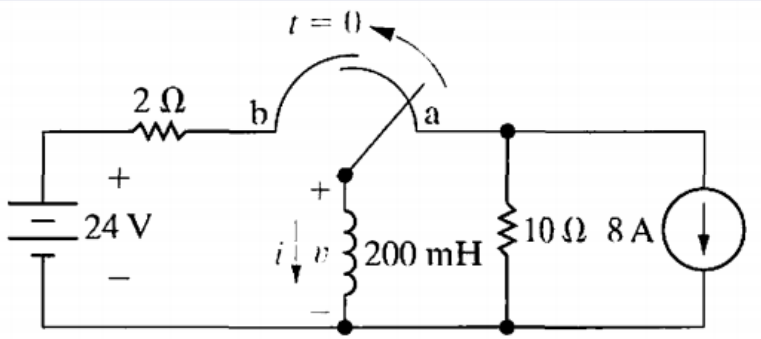
Figure 7.24 ▲ Four possible first-order circuits.
(a) An inductor connected to a Thévenin equivalent.
(b) An inductor connected to a Norton equivalent.
(c) A capacitor connected to a Thévenin equivalent.
(d) A capacitor connected to a Norton equivalent.

The Step and the Total Responses of RL and RC Circuits

4

- Example-5: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.

a) Find the expression for $i(t)$ for $t \geq 0$



Solution:

a) The switch has been in position a for a long time, so the 200 mH inductor is a short circuit across the 8 A current source. Therefore, the inductor carries an initial current of 8 A. This current is oriented opposite to the reference direction for i, thus I_0 is -8 A.

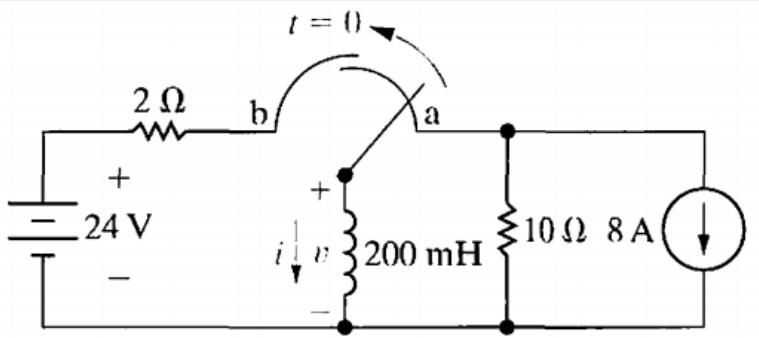
When the switch is in position b, the final value of i will be 24/2=12 A. The time constant of the circuit is 200/2=100 ms. Using these values,

$$i(t) = 12 + (-8 - 12)e^{-t/0.1} = 12 - 20e^{-10t}\ A, \qquad t \geq 0$$

The Step and the Total Responses of RL and RC Circuits

4

- Example-5: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.



b) What is the initial voltage across the inductor just after the switch has been moved to position b?

Solution:

b) The voltage across the inductor is

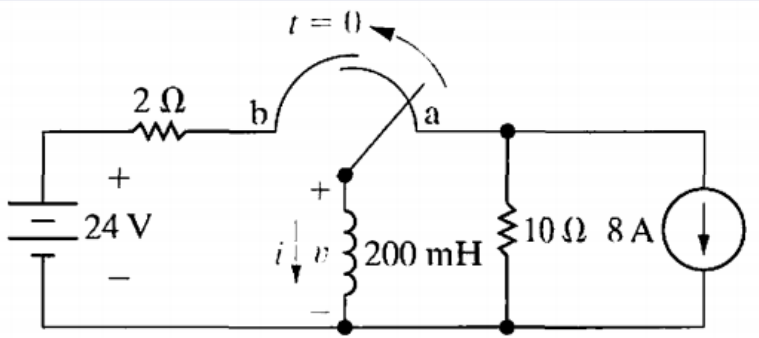
$$v(t) = L\frac{di}{dt} = 0.2(200e^{-10t}) = 40e^{-10t}\ \text{V}, \qquad t \geq 0^{+}$$

The initial inductor voltage is

$$v(0^{+}) = 40\ V$$

The Step and the Total Responses of RL and RC Circuits

- Example-5: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.



c) How many milliseconds after the switch has been moved does the inductor voltage equal 24 V?

d) Find the time at which the inductor voltage equals 24 V.

Solution: c) In the instant after the switch has been moved to position b, the inductor sustains a current of 8 A counterclokwise around the newly formed closed path. This current causes a 16 V drop across the 2 Ω resistor. This voltage drop adds to the drop across the source, producing a 40 V drop across the inductor.

d) We find the time at which the inductor voltage equals 24 V by solving the expression

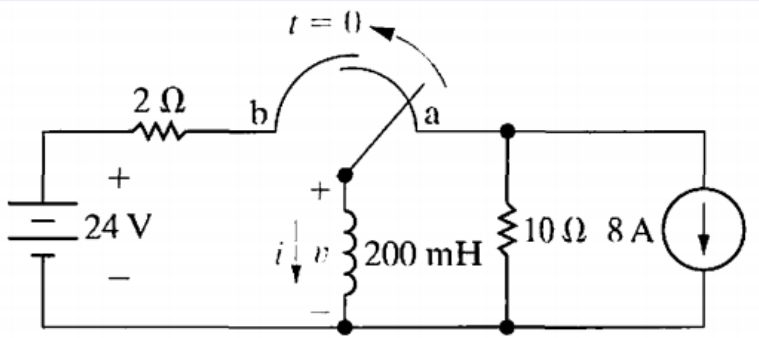
$$24 = 40e^{-10t} \quad \text{for } t: \quad t = \frac{1}{10} ln \frac{40}{24} = 51.08 \times 10^{-3} = 51.08\, ms$$

The Step and the Total Responses of RL and RC Circuits

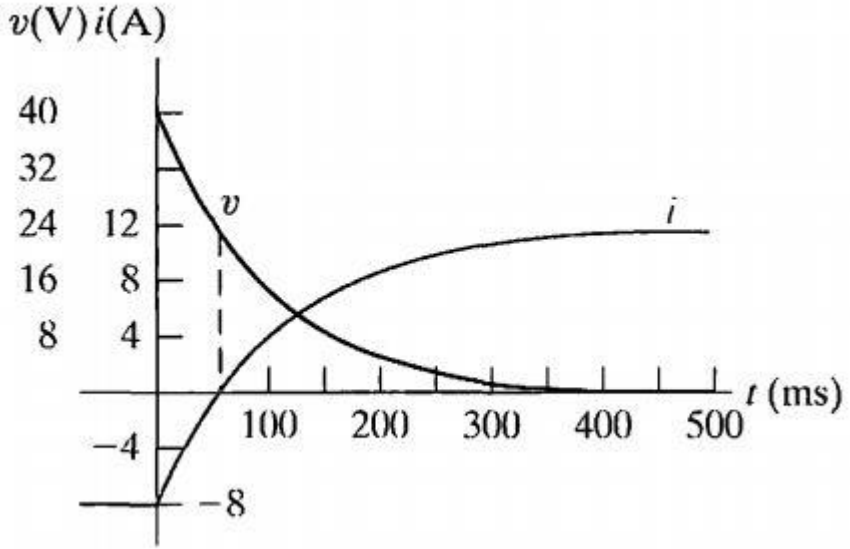
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- Example-5: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.

e) Plot both $i(t)$ and $v(t)$ versus t.



Solution: e) Note that the time when the current equals zero corresponds to the time when the inductor voltage equals the source voltage of 24 V, as predicted by KVL.



End of the Lesson (18.12.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist